ENGR 102 Summer Bridge
Day 6 Student Quiz

Multi-Step Calculations, Summaries, and Conditional Output

20 points • approximately 20-25 minutes

Name		UIN		Section	

Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Evaluate ceiling-style kit arithmetic. 4 points

```
groups = 7
items_per_group = 5
spares = 4
items = groups * items_per_group + spares
kits = (items + 11) // 12
leftover_space = kits * 12 - items
print(kits, leftover_space)
```

Show the item total, adjusted numerator, quotient, and exact output.

Question 2. Trace a numeric summary and formatting boundary. 4 points

```
values = [4.5, 7.0, 2.5, 6.0]
total = sum(values)
average = total / len(values)
print(min(values), max(values), f"{average:.2f}")
```

Compute every summary and give exact output, including the formatted average.

Question 3. Combine arithmetic with a conditional label. 4 points

```
mass = 18.0
purity = 84.5
cost = 12 + 1.8 * mass
hold = purity < 85 or mass > 25

if hold:
    label = "hold"
else:
    label = "accepted"

print(f"{label}: {cost:.2f}")
```

Show the cost calculation and each Boolean component before giving exact output.

Question 4. Program construction — 8 points

Write one complete Python program that satisfies every requirement.

- Read length, width, and height as floats.
- Print invalid if any dimension is not positive.
- Otherwise compute volume and rectangular-prism surface area.
- Classify oversize when volume is above 500 or the largest dimension is above 12; otherwise classify standard.
- Print Volume and Surface area to two decimals, followed by Class: and the label.

Test evidence / expected result